

Introduction to group representation and Maschke's theorem

Filippo Rossi

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We shall always write $\mathbb{K}$ for an arbitrary field
and V for an arbitrary vector space of dimension n defined on $\mathbb{K}$.

1 Matrix representation

Informally, a representation of a group is a way of writing it down as a group of matrices.

Example 1.1. Consider C_4 (i.e., $\mathbb{Z}/4\mathbb{Z}$):

$$C_4 = \langle \mu \mid \mu^4 = e \rangle$$

Define:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

It is easy to see that $M^4 = I$, then we can build the following map:

$$\begin{array}{ccc} \phi: C_4 & \rightarrow & GL_2(\mathbb{R}) \\ \mu & \mapsto & M \end{array}$$

Moreover, we have that ϕ is an isomorphism between C_4 and $\langle M \rangle$.

Definition 1.2. Let G be a group. A *matrix representation* of G is a homomorphism

$$\rho : G \rightarrow GL_n(\mathbb{K})$$

where $n \in \mathbb{N}$ and is called the dimension of the matrix representation.

Remark 1.3. In the example 1.1 the representation is also injective, but in general we do not require this property to be satisfied.

Definition 1.4. A matrix representation ρ is *faithful* if ρ is injective.

Example 1.5. Let G be any group, and $n \in \mathbb{N}$. Define

$$\begin{aligned} \rho : G &\rightarrow GL_n(\mathbb{K}) \\ g &\mapsto I_n \end{aligned}$$

This is a matrix representation since ρ is the trivial homomorphism. This is known as the trivial matrix representation of G of dimension n .

Now, we want to define equivalence relation between matrix representations. In order to do this:

Definition 1.6. Let $P \in GL_n(\mathbb{K})$ an invertible matrix. The following map

$$\begin{aligned} c_P : GL_n(\mathbb{K}) &\rightarrow GL_n(\mathbb{K}) \\ M &\mapsto P^{-1}MP \end{aligned}$$

is called the *conjugate by P* map.

Definition 1.7. Two matrix representations of G

$$\rho_1 : G \rightarrow GL_n(\mathbb{K}) \quad \rho_2 : G \rightarrow GL_n(\mathbb{K})$$

are *equivalent* if exists $P \in GL_n(\mathbb{K})$ such that $\rho_2 = c_P \circ \rho_1$

It's easy to show that the relation above is an equivalence relation.

2 Linear representation

Now we want to give an other definition of representation, using the vector space on the field $\mathbb{K}$.

Definition 2.1. A *linear representation* (or just a *representation*) of a group G is a choice of an n -dimensional vector space V on $\mathbb{K}$ and a homomorphism

$$\rho : G \rightarrow GL(V)$$

Are there any differences between a representation and a matrix representation?

Remark 2.2. If we pick a base B for V , we have that $GL(V) \xrightarrow{\psi_B} GL_n(\mathbb{K})$. Then, if $\underline{Rep_{\mathbb{K}}(G)}$ is the set of the matrix representation and $\underline{Rep_{\mathbb{K},V}(G)}$ is the set of representation on V , then we have the following one-to-one correspondence:

$$\eta : \frac{\underline{Rep_{\mathbb{K},V}(G)}}{\rho} \rightarrow \frac{\underline{Rep_{\mathbb{K}}(G)}}{\psi_B \circ \rho}$$

Then, fixed a vector space and a basis, we can see a representation as a matrix representation, and vice versa.

Definition 2.3. Let be V a vector space, A a basis for V and ρ a representation. We can define the *matrix representation associated to A* in the following way:

$$\begin{aligned} \rho^A : G &\rightarrow GL_n(K) \\ g &\mapsto \rho^A(g) \end{aligned}$$

where $\rho^A(g)$ is the linear map $\rho(g)$ written as a matrix with respect to the base A .

Lemma 2.4. Let be V a vector space, $\rho : G \rightarrow GL(V)$ be a representation of a group G and A, B two basis for V . Then the matrix representations ρ_A and ρ_B associated to A and B are equivalent.

One can also prove that two equivalent matrix representation come from the same representation, then we have that two matrix representation are equivalent if and only if they describe the same representation in different bases.

3 Regular representation

Suppose we have a subgroup $G \subset S_n$, then we can write down an n -dimensional representation of G in the following way.

Let V be an n -dimensional vector space with a basis $\{v_1, \dots, v_n\}$. Since every element $g \in G$ is a permutation of the set $\{1, \dots, n\}$ we can define the following map:

$$\begin{aligned} \rho(g) : V &\longrightarrow V \\ \sum_{i=1}^n a_i v_i &\longmapsto \sum_{i=1}^n a_i v_{g(i)} \end{aligned}$$

i.e. the map that sends the i -th element of the base in the $g(i)$ -th element of the base; clearly $\rho(g)$ is a bijective linear map between vector space for all $g \in G$. We also have that ρ is linear, indeed

$$\begin{aligned} \rho(g) \circ \rho(h) : V &\longrightarrow V \\ \sum_{i=1}^n a_i v_i &\longmapsto \sum_{i=1}^n a_i v_{g(h(i))} = \sum_{i=1}^n a_i v_{(gh)(i)} \end{aligned}$$

then we have that

$$\rho(g) \circ \rho(h) = \rho(gh)$$

Therefore, we have that ρ is a homomorphism. Moreover, we know that each finite group such that $|G| = n$ for some $n \in \mathbb{N}$ can be embedded in S_n , then we can define the previous homomorphism for each finite group:

Definition 3.1. Let G be a finite group such that $|G| = n$, the previous homomorphism together with any n -dimensional vector space V on $\mathbb{K}$ is a representation of G and is called the *regular representation*.

4 Maschke's theorem

Definition 4.1. A *subrepresentation* of a representation $\rho : G \rightarrow GL_n(V)$ with G a group and V a vector space is a vector subspace $W \subset V$ such that

$$\rho(g)(x) \in W \quad \forall g \in G \text{ and } x \in W$$

Definition 4.2. Let V be a vector space, G a group, ρ a representation and $W \subset V$ a subrepresentation. If exists another subrepresentation $U \subset V$ such that $U \cap W = \{0\}$ and $\dim V = \dim U + \dim W$, as we know we have that $V = U \oplus W$ and such a U is called a *complementary subrepresentation* to W .

Do they always exists? The *Maschke's theorem* answers to this question.

Theorem 4.3. (Maschke's theorem) Let $\rho : G \rightarrow GL(V)$ be a representation, and let $W \subset V$ be a subrepresentation. Then there exists a complementary subrepresentation $U \subset V$ to W .

Definition 4.4. If $\rho : G \rightarrow GL(V)$ is a representation with no subrepresentations (apart from the trivial subrepresentations $0 \subset V$ and $V \subset V$) then we call it an irreducible representation.

The real power of Maschke's theorem is the following Corollary:

Corollary 4.5. If ρ and V is a representation for G , then we have that

$$V = \bigoplus_{i=1}^r U_i$$

where each U_i is an irreducible subrepresentations.

Proof. Let V be a representation of G , of dimension n . If V is irreducible, we're done. If not, V contains a non trivial subrepresentation $W \subset V$, and by Maschke's Theorem,

$$V = W \oplus U$$

for some other subrepresentation U . Both W and U have dimension less than n . If they're both irreducible, we're done. If not, one of them contains a non trivial subrepresentation, so it splits as a direct sum of smaller subrepresentations. Since n is finite, this process will terminate in a finite number of steps. $\square$

So every representation is built up from irreducible representations in a straight-forward way. This makes irreducible representations very important, so we abbreviate the name and call them irreps. They're like the 'prime numbers' or 'primary ideals' of representation theory.

Example 4.6. We want to study the regular representation of $C_3 = \langle \eta \mid \eta^3 = 3 \rangle$ with $\mathbb{K}^3$ as vector space in the canonical base $B := \{e_1, e_2, e_3\}$. First of all, $C_3 \stackrel{f}{\cong} A_3 < S_3$, where $f(\eta) := (123)$. Then, instead of working with C_3 , we can work with A_3 . We have that

$$\rho_{\text{reg}}((123))(e_1a_1 + e_2a_2 + e_3a_3) = e_1a_3 + e_2a_1 + e_3a_2$$

$$\rho_{\text{reg}}((132))(e_1a_1 + e_2a_2 + e_3a_3) = e_1a_2 + e_2a_3 + e_3a_1$$

Now we want to write the representation as a matrix representation respect to the canonical base, i.e.

$$\rho_{\text{reg}}^B((123)) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\rho_{\text{reg}}^B((132)) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

We have just wrote the matrices in the right way, indeed

$$\rho_{\text{reg}}^B((123)) \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} a_3 \\ a_1 \\ a_2 \end{pmatrix} = \rho_{\text{reg}}((123)) \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

$$\rho_{\text{reg}}^B((132)) \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} a_2 \\ a_3 \\ a_1 \end{pmatrix} = \rho_{\text{reg}}((132)) \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

Suppose now that $x \in \mathbb{K}^3$ is an eigenvector of $\rho_{\text{reg}}^B(\mu)$ where $\mu := (123)$, then exists $a \in \mathbb{K}$ such that $\rho_{\text{reg}}^B(\mu)x = ax$. We also have that

$$\rho_{\text{reg}}^B(\mu^2)(x) = \rho_{\text{reg}}^B(\mu)(\rho_{\text{reg}}^B(\mu)x) = \rho_{\text{reg}}^B(\mu)(ax) = a^2x$$

then x is also an eigenvector of $\rho_{\text{reg}}^B(\mu^2)$.

Now let $g \in A_3$ and $l \in \langle x \rangle$, then exists $\lambda \in \mathbb{K}$ such that $l = \lambda x$. Moreover, from the previous computation, follows that

$$\rho_{\text{reg}}^B(g)(l) = \rho_{\text{reg}}^B(g)(\lambda x) = {}^1\lambda \rho_{\text{reg}}^B(g)(x) = \begin{cases} (\lambda a)x \in \langle x \rangle, & \text{if } g = \mu \\ (\lambda a^2)x \in \langle x \rangle, & \text{if } g = \mu^2 \end{cases}$$

then we have that $\langle x \rangle$ is a 1-dimensional subrepresentation. The eigenvectors of $\rho_{\text{reg}}^B(\mu)$ are:

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (\lambda_1 = 1) \quad \begin{pmatrix} 1 \\ \omega^{-1} \\ \omega \end{pmatrix} \quad (\lambda_2 = \omega) \quad \begin{pmatrix} 1 \\ \omega \\ \omega^{-1} \end{pmatrix} \quad (\lambda_3 = \omega^{-1})$$

So ρ_{reg} is the direct sum of 3 one-dimensional irreps:

$$U_1 = \left\langle \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\rangle \quad U_2 = \left\langle \begin{pmatrix} 1 \\ \omega^{-1} \\ \omega \end{pmatrix} \right\rangle \quad U_3 = \left\langle \begin{pmatrix} 1 \\ \omega \\ \omega^{-1} \end{pmatrix} \right\rangle$$

¹This equality holds since $\rho_{\text{reg}}^B(g)$ is a linear map between vector space.

In the previous example, each subrepresentation corresponds to the vector space generated by an eigenvector: does this always happen?

Proposition 4.7. Let $\rho : G \rightarrow GL_n(\mathbb{K})$ be a matrix representation. Then there exists a basis of $\mathbb{K}^n$ in which every matrix $\rho(g)$ is diagonal if and only if ρ is a direct sum of 1-dimensional irreps.

Proof.

($\Rightarrow$) Let $\{x_1, \dots, x_n\}$ be such a basis. Then x_i is an eigenvector for every $\rho(g)$, so, as we have done in the previous example, $\langle x_i \rangle$ is a 1-dimensional subrepresentation, and

$$\mathbb{K}^n = \langle x_1 \rangle \oplus \langle x_2 \rangle \oplus \dots \oplus \langle x_n \rangle$$

($\Leftarrow$) Suppose $\mathbb{K}^n = U_1 \oplus \dots \oplus U_n$ with each U_i a 1-dimensional subrepresentation. Pick a (non-zero) vector x_i from each U_i . Then $\{x_1, \dots, x_n\}$ is a basis for $\mathbb{K}^n$. For any $g \in G$, the matrix $\rho(g)$ preserves $\langle x_i \rangle = U_i$ for all i , so $\rho(g)$ is a diagonal matrix with respect to this basis. $\square$

Bibliography

Course material from lectures of the course *Group Representation Theory (2014)* in *UCL Department of Mathematics* holded by "Ed Segal".

Link to notes: <https://shorturl.at/W0Za2>